

HSCTF 2020 Crypto - Affina and the quadratics

deuterium — 2020-06-06

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Abstract

CTF writeup: Quadratic homework moonlights as an affine cipher in HSCTF 2020's Affina and the Quadratics. Its coefficients give the game away.

Affina and the Quadratics

Challenge

65 Solves



Affina and the Quadratics

428

crypto

new

Affina was struggling with her graphing quadratics homework. Bored, she decided to cheat by using Desmos to graph the given quadratic, and then realized that she could use it to send messages to her best friend without anyone noticing!

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Note: Affina uses a 26 character charset and the numbers are encoded differently.



affffinaaaa-p...

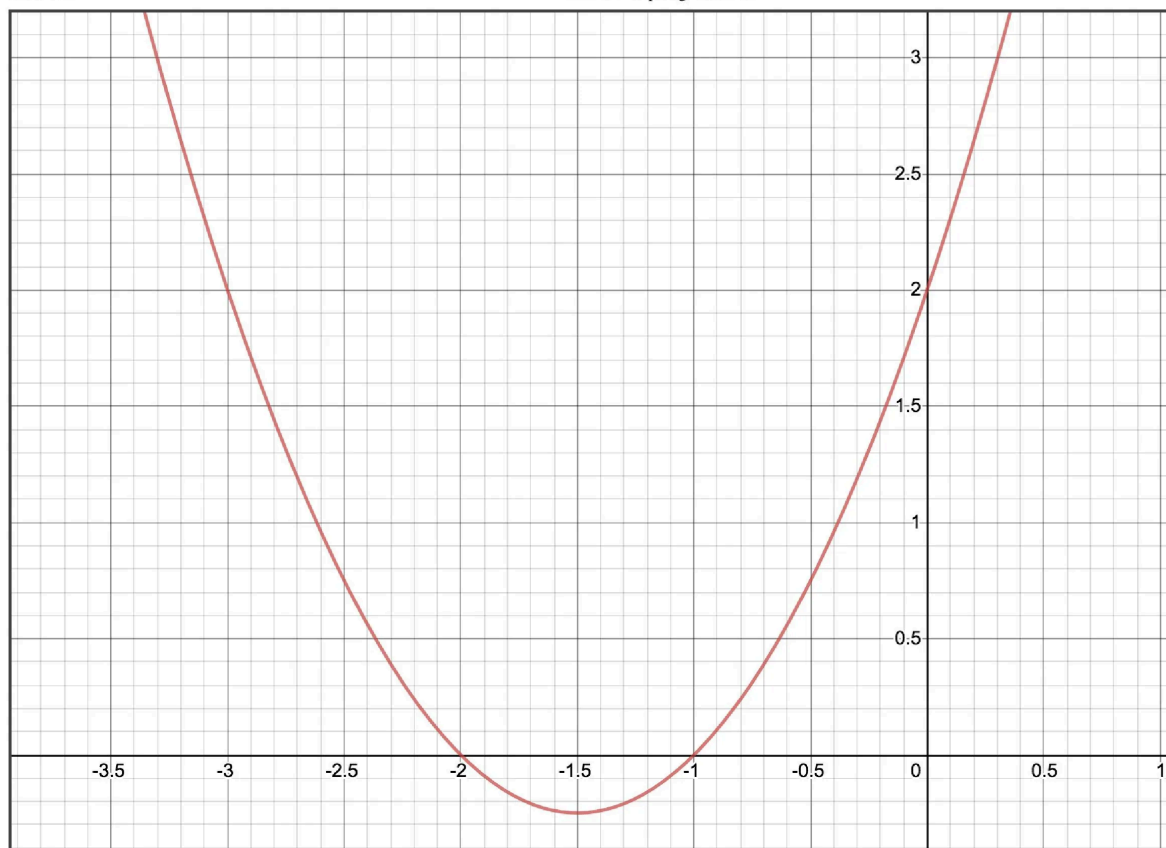


Affina_and_t...

Flag

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We are given this image, and hinted towards the name Affina.



With a description reading

Affina was struggling with her graphing quadratics homework. Bored, she decided to cheat by using Desmos to graph the given quadratic, and then realized that she could use it to send messages to her best friend without anyone noticing!

She sent the following message to her friend: 7rr4p6_4e_4ph6bo8hap2?

Can you decrypt it using the image of the quadratic Affina used? Note: the flag should make relative sense.

The task is to GUESS how Affine cipher is used with respect to the quadratic $(x+1)*(x+2) = 0$

After having many GUESSES, I GUESSED that it could be simple affine cipher with mapping $3x+2 \bmod 26$ as $(x+1)*(x+2) = x^2 + 3x + 2$ as hey!, why should it be quadratic right?

But dont we have letters too? So I GUESSED we shouldnt care about them for now. What about underscores? GUESS that they are just word-delimiters.

```
c = 3x + 2 mod 26
c - 2 = 3x mod 26
3^-1 * (c - 2) = x mod 26
9c - 18 = x mod 26
x = 9c + 8 mod 26
```

```
ct = "7rr4p6_4e_4ph6bo8hap2"
alpha = "rrpephbohpa"
ct26 = [ chr(i) - 97 for i in alpha]
pt = [ chr((9*c + 8)%26 + 97) for c in ct26 ]
```

putting it in places, we get 7ff4n6_4s_4nt6re8tin2

Now why GUESS that number are encoded with the same affine cipher modulo when you can easily GUESS the flag is 4ff1n3_1s_1nt3re5tin6 😊

Interactive controls are omitted from this PDF. See the canonical HTML version.